

Titration report



Method

Method ID	CDC_M003	Release state	Released
Name	Titration with HCl - Liquid	Modified by	PredefinedUser
Modified on	2/25/2026 5:34:04 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.012966 mL
		Concentration R2	0.10026 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.007864 mL
		Concentration R2	0.10016 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.242328 mL
		Concentration R2	0.10485 mol/L

Statistics

R1

Mean	5.087719 mL	Minimum	5.007864 mL
Standard deviation	0.133919 mL	Rel. standard deviation	2.632 %
Total samples	3	Samples excluded	0

R2

Mean	0.10176 mol/L	Minimum	0.10016 mol/L
Standard deviation	0.00268 mol/L	Rel. standard deviation	2.633 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165302	Task name	Unused DG111 INC=0.02 mL, VDISP = 2 mL
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.012966 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10026 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.007864 mL	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10016 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.242328 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10485 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/2/2026 8:22:39 AM
Overall state	OK
Sample start time	3/2/2026 8:22:39 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/2/2026 8:22:44 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/2/2026 8:22:44 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:15 min

Stir (Stir1)

Creation time	3/2/2026 8:23:13 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	3/2/2026 8:25:14 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	102.2 mV
EHNv1	-76.0 mV
Q1	0.501297 mmol
VEQ1	5.012966 mL
EST	-179.3 mV
VEX	0.187034 mL
VEND	5.2000 mL
t	09:11 min
QEND	0.52000 mmol
QEX	0.018703 mmol
EPD	-86.4 mV
nEQ	1

Consumption (R1)

Creation time	3/2/2026 8:34:32 AM
Overall state	OK
Result	5.012966 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	3/2/2026 8:34:32 AM
Overall state	OK
Result	0.10026 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/2/2026 8:34:55 AM
Overall state	OK
Sample start time	3/2/2026 8:34:55 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/2/2026 8:34:57 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/2/2026 8:34:57 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	3/2/2026 8:35:24 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	3/2/2026 8:37:25 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	98.5 mV
EHNv1	-75.6 mV
Q1	0.500786 mmol

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VEQ1	5.007864 mL
EST	-178.1 mV
VEX	0.192136 mL
VEND	5.2000 mL
t	09:05 min
QEND	0.52000 mmol
QEX	0.019214 mmol
EPD	-86.0 mV
nEQ	1

Consumption (R1)

Creation time	3/2/2026 8:46:34 AM
Overall state	OK
Result	5.007864 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	3/2/2026 8:46:35 AM
Overall state	OK
Result	0.10016 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/2/2026 8:47:09 AM
Overall state	OK
Sample start time	3/2/2026 8:47:09 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/2/2026 8:47:11 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/2/2026 8:47:12 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

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Creation time 3/2/2026 8:47:39 AM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 3/2/2026 8:49:39 AM
Overall state OK
Terminated at Number of EQPs
EEQ1 109.6 mV
EHN1 -74.7 mV
Q1 0.524233 mmol
VEQ1 5.242328 mL
EST -180.5 mV
VEX 0.197672 mL
VEND 5.4400 mL
t 09:40 min
QEND 0.54400 mmol
QEX 0.019767 mmol
EPD -87.2 mV
nEQ 1

Consumption (R1)

Creation time 3/2/2026 8:59:24 AM
Overall state OK
Result 5.242328 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

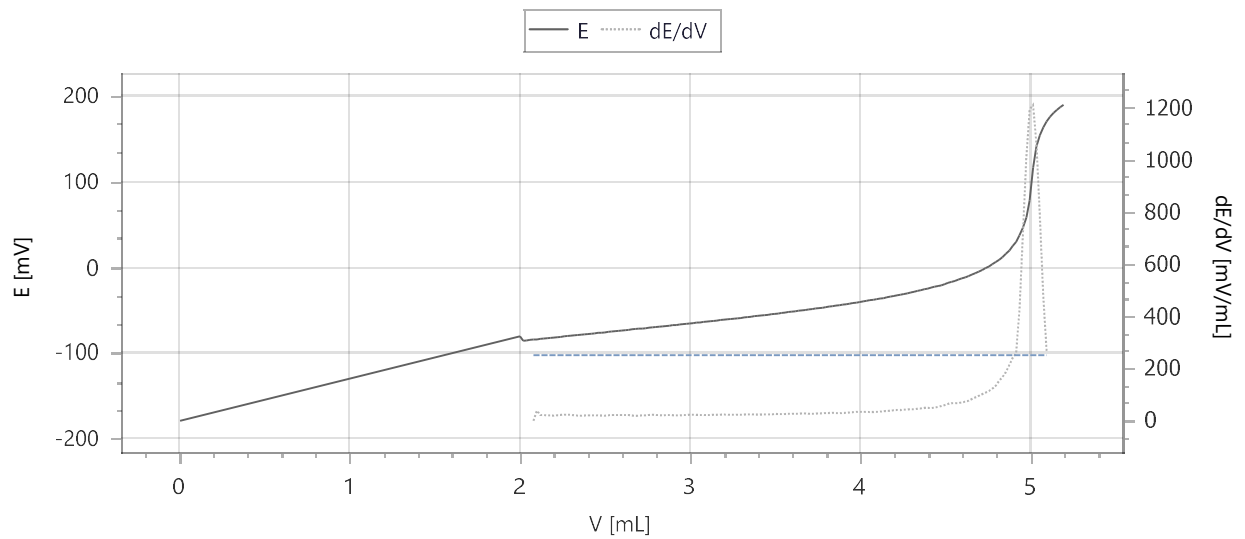
Creation time 3/2/2026 8:59:25 AM
Overall state OK
Result 0.10485 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-179.3		25.0			
6	2.0000	-80.8	2.0000	25.0	98.5		
40	2.0200	-86.0	0.0200	25.0	-5.2		
42	2.0400	-85.5	0.0200	25.0	0.5		
46	2.0600	-85.1	0.0200	25.0	0.4		
49	2.0800	-84.6	0.0200	25.0	0.5	-0.97	
52	2.1000	-84.4	0.0200	25.0	0.2	37.80	
55	2.1200	-83.8	0.0200	25.0	0.6	21.08	
58	2.1400	-83.3	0.0200	25.0	0.5	21.26	
61	2.1600	-83.0	0.0200	25.0	0.3	19.88	
64	2.1800	-82.6	0.0200	25.0	0.4	19.56	
67	2.2000	-82.2	0.0200	25.0	0.4	18.92	
70	2.2200	-81.8	0.0200	25.0	0.4	20.14	
73	2.2400	-81.5	0.0200	25.0	0.3	21.57	
76	2.2600	-80.9	0.0200	25.0	0.6	22.38	
79	2.2800	-80.5	0.0200	25.0	0.4	22.49	
82	2.3000	-80.1	0.0200	25.0	0.4	22.09	
85	2.3200	-79.6	0.0200	25.0	0.5	20.28	
88	2.3400	-79.2	0.0200	25.0	0.4	18.61	
91	2.3600	-78.9	0.0200	25.0	0.3	18.44	
94	2.3800	-78.5	0.0200	25.0	0.4	19.28	
98	2.4000	-78.2	0.0200	25.0	0.3	19.72	
100	2.4200	-77.7	0.0200	25.0	0.5	19.98	
104	2.4400	-77.3	0.0200	25.0	0.4	20.35	
107	2.4600	-76.8	0.0200	25.0	0.5	19.46	
110	2.4800	-76.6	0.0200	25.0	0.2	19.19	
113	2.5000	-76.2	0.0200	25.0	0.4	18.73	
116	2.5200	-75.7	0.0200	25.0	0.5	20.26	
119	2.5400	-75.4	0.0200	25.0	0.3	21.24	
122	2.5600	-74.9	0.0200	25.0	0.5	21.58	
125	2.5800	-74.5	0.0200	25.0	0.4	20.76	

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128	2.6000	-74.0	0.0200	25.0	0.5	20.74
131	2.6200	-73.7	0.0200	25.0	0.3	21.29
134	2.6400	-73.3	0.0200	25.0	0.4	21.39
137	2.6600	-72.8	0.0200	25.0	0.5	19.11
140	2.6800	-72.4	0.0200	25.0	0.4	17.87
143	2.7000	-72.0	0.0200	25.0	0.4	18.85
146	2.7200	-71.7	0.0200	25.0	0.3	19.65
149	2.7400	-71.6	0.0200	25.0	0.1	20.20
152	2.7600	-70.8	0.0200	25.0	0.8	21.41
156	2.7800	-70.3	0.0200	25.0	0.5	21.98
158	2.8000	-70.0	0.0200	25.0	0.3	21.49
162	2.8200	-69.7	0.0200	25.0	0.3	20.65
165	2.8400	-69.2	0.0200	25.0	0.5	19.12
168	2.8600	-68.8	0.0200	25.0	0.4	20.69
171	2.8800	-68.4	0.0200	25.0	0.4	20.80
174	2.9000	-67.9	0.0200	25.0	0.5	20.57
177	2.9200	-67.6	0.0200	25.0	0.3	19.65
180	2.9400	-67.2	0.0200	25.0	0.4	19.41
183	2.9600	-66.8	0.0200	25.0	0.4	20.85
186	2.9800	-66.3	0.0200	25.0	0.5	21.46
189	3.0000	-66.0	0.0200	25.0	0.3	21.97
192	3.0200	-65.5	0.0200	25.0	0.5	22.17
195	3.0400	-64.9	0.0200	25.0	0.6	21.71
198	3.0600	-64.7	0.0200	25.0	0.2	21.42
201	3.0800	-64.2	0.0200	25.0	0.5	21.10
204	3.1000	-63.7	0.0200	25.0	0.5	20.67
207	3.1200	-63.4	0.0200	25.0	0.3	21.16
210	3.1400	-62.9	0.0200	25.0	0.5	22.37
214	3.1600	-62.5	0.0200	25.0	0.4	21.70
217	3.1800	-62.0	0.0200	25.0	0.5	22.51
220	3.2000	-61.6	0.0200	25.0	0.4	23.08
223	3.2200	-61.1	0.0200	25.0	0.5	22.21
226	3.2400	-60.7	0.0200	25.0	0.4	21.98
229	3.2600	-60.2	0.0200	25.0	0.5	22.01
232	3.2800	-59.8	0.0200	25.0	0.4	22.21
235	3.3000	-59.4	0.0200	25.0	0.4	22.21
238	3.3200	-58.9	0.0200	25.0	0.5	22.78
241	3.3400	-58.4	0.0200	25.0	0.5	23.79
244	3.3600	-58.0	0.0200	25.0	0.4	24.02
247	3.3800	-57.5	0.0200	25.0	0.5	22.98
250	3.4000	-57.0	0.0200	25.0	0.5	22.78
253	3.4200	-56.5	0.0200	25.0	0.5	22.95
256	3.4400	-56.2	0.0200	25.0	0.3	23.21
259	3.4600	-55.7	0.0200	25.0	0.5	23.29
262	3.4800	-55.1	0.0200	25.0	0.6	23.69
265	3.5000	-54.7	0.0200	25.0	0.4	25.30
268	3.5200	-54.2	0.0200	25.0	0.5	25.03
272	3.5400	-53.7	0.0200	25.0	0.5	24.45
275	3.5600	-53.2	0.0200	25.0	0.5	25.02
278	3.5800	-52.6	0.0200	25.0	0.6	26.00
281	3.6000	-52.3	0.0200	25.0	0.3	25.71
284	3.6200	-51.6	0.0200	25.0	0.7	26.08

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287	3.6400	-51.1	0.0200	25.0	0.5	26.19
290	3.6600	-50.6	0.0200	25.0	0.5	27.03
293	3.6800	-50.1	0.0200	25.0	0.5	27.37
296	3.7000	-49.5	0.0200	25.0	0.6	25.61
299	3.7200	-49.0	0.0200	25.0	0.5	25.98
302	3.7400	-48.4	0.0200	25.0	0.6	26.53
305	3.7600	-48.0	0.0200	25.0	0.4	26.93
308	3.7800	-47.5	0.0200	25.0	0.5	27.79
311	3.8000	-46.8	0.0200	25.0	0.7	28.27
314	3.8200	-46.2	0.0200	25.0	0.6	28.99
317	3.8400	-45.7	0.0200	25.0	0.5	29.33
320	3.8600	-45.1	0.0200	25.0	0.6	28.49
323	3.8800	-44.6	0.0200	25.0	0.5	27.91
326	3.9000	-43.9	0.0200	25.0	0.7	28.71
330	3.9200	-43.4	0.0200	25.0	0.5	30.00
333	3.9400	-42.8	0.0200	25.0	0.6	30.74
336	3.9600	-42.2	0.0200	25.0	0.6	31.76
339	3.9800	-41.5	0.0200	25.0	0.7	33.08
342	4.0000	-40.8	0.0200	25.0	0.7	33.51
345	4.0200	-40.2	0.0200	25.0	0.6	32.98
348	4.0400	-39.5	0.0200	25.0	0.7	33.39
351	4.0600	-38.8	0.0200	25.0	0.7	32.34
354	4.0800	-38.3	0.0200	25.0	0.5	32.69
357	4.1000	-37.5	0.0200	25.0	0.8	33.31
360	4.1200	-36.8	0.0200	25.0	0.7	34.40
363	4.1400	-36.3	0.0200	25.0	0.5	35.45
366	4.1600	-35.4	0.0200	25.0	0.9	36.85
369	4.1800	-34.7	0.0200	25.0	0.7	37.70
372	4.2000	-33.9	0.0200	25.0	0.8	39.24
375	4.2200	-33.2	0.0200	25.0	0.7	40.35
378	4.2400	-32.3	0.0200	25.0	0.9	39.93
381	4.2600	-31.5	0.0200	25.0	0.8	41.77
384	4.2800	-30.7	0.0200	25.0	0.8	42.24
388	4.3000	-29.8	0.0200	25.0	0.9	43.27
391	4.3200	-29.0	0.0200	25.0	0.8	43.42
394	4.3400	-28.0	0.0200	25.0	1.0	44.99
397	4.3600	-27.2	0.0200	25.0	0.8	46.50
400	4.3800	-26.2	0.0200	25.0	1.0	48.60
403	4.4000	-25.3	0.0200	25.0	0.9	48.70
406	4.4200	-24.2	0.0200	25.0	1.1	48.09
409	4.4400	-23.2	0.0200	25.0	1.0	50.00
412	4.4600	-22.2	0.0200	25.0	1.0	52.10
415	4.4800	-21.4	0.0200	25.0	0.8	55.41
418	4.5000	-20.1	0.0200	25.0	1.3	59.35
421	4.5200	-18.7	0.0200	25.0	1.4	63.56
424	4.5400	-17.5	0.0200	25.0	1.2	66.32
427	4.5600	-16.2	0.0200	25.0	1.3	66.30
430	4.5800	-14.8	0.0200	25.0	1.4	66.67
433	4.6000	-13.5	0.0200	25.0	1.3	69.09
436	4.6200	-12.1	0.0200	25.0	1.4	72.03
439	4.6400	-10.7	0.0200	25.0	1.4	75.60
442	4.6600	-8.9	0.0200	25.0	1.8	82.51

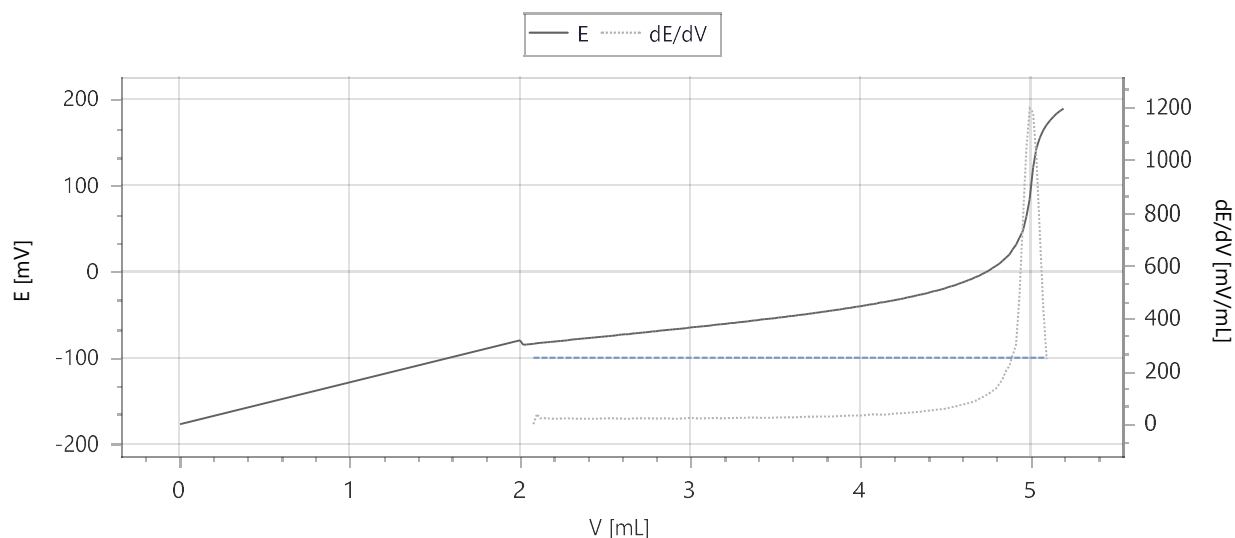
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446	4.6800	-7.4	0.0200	25.0	1.5	88.57	
449	4.7000	-5.6	0.0200	25.0	1.8	94.19	
452	4.7200	-3.5	0.0200	25.0	2.1	100.55	
455	4.7400	-1.3	0.0200	25.0	2.2	106.98	
458	4.7600	0.6	0.0200	25.0	1.9	112.52	
461	4.7800	3.1	0.0200	25.0	2.5	121.78	
464	4.8000	5.7	0.0200	25.0	2.6	132.50	
467	4.8200	8.4	0.0200	25.0	2.7	151.48	
470	4.8400	11.4	0.0200	25.0	3.0	166.57	
473	4.8600	15.5	0.0200	25.0	4.1	185.03	
476	4.8800	19.2	0.0200	25.0	3.7	212.13	
479	4.9000	24.3	0.0200	25.0	5.1	239.54	
482	4.9200	29.0	0.0200	25.0	4.7	257.68	
485	4.9400	36.6	0.0200	25.0	7.6	422.54	
489	4.9600	45.6	0.0200	25.0	9.0	712.81	
494	4.9800	57.6	0.0200	25.0	12.0	1001.64	
498	5.0000	77.4	0.0200	25.0	19.8	1189.11	
	5.012966	102.2				1210.72	EQP
507	5.0200	115.7	0.0200	25.0	38.3	1205.93	
511	5.0400	140.0	0.0200	25.0	24.3	1055.66	
514	5.0600	153.2	0.0200	25.0	13.2	778.76	
518	5.0800	162.6	0.0200	25.0	9.4	473.11	
521	5.1000	169.5	0.0200	25.0	6.9	261.78	
524	5.1200	174.9	0.0200	25.0	5.4		
527	5.1400	179.3	0.0200	25.0	4.4		
530	5.1600	183.0	0.0200	25.0	3.7		
534	5.1800	186.1	0.0200	25.0	3.1		
536	5.2000	188.7	0.0200	25.0	2.6		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time	V	Meas. value	dV	Temp.	dE	dE/dV	EQP
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[s]	[mL]	[mV]	[mL]	[°C]	[mV]	[mV/mL]
0	0.0000	-178.1		25.0		
6	2.0000	-80.2	2.0000	25.0	97.9	
40	2.0200	-85.5	0.0200	25.0	-5.3	
42	2.0400	-85.0	0.0200	25.0	0.5	
46	2.0600	-84.6	0.0200	25.0	0.4	
49	2.0800	-84.2	0.0200	25.0	0.4	-1.63
52	2.1000	-83.8	0.0200	25.0	0.4	37.60
55	2.1200	-83.4	0.0200	25.0	0.4	21.00
58	2.1400	-82.9	0.0200	25.0	0.5	21.58
61	2.1600	-82.5	0.0200	25.0	0.4	20.71
64	2.1800	-82.1	0.0200	25.0	0.4	19.99
67	2.2000	-81.7	0.0200	25.0	0.4	19.51
70	2.2200	-81.3	0.0200	25.0	0.4	19.42
73	2.2400	-81.0	0.0200	25.0	0.3	20.00
76	2.2600	-80.5	0.0200	25.0	0.5	20.29
79	2.2800	-80.1	0.0200	25.0	0.4	20.49
82	2.3000	-79.7	0.0200	25.0	0.4	20.81
85	2.3200	-79.3	0.0200	25.0	0.4	20.00
88	2.3400	-78.9	0.0200	25.0	0.4	19.19
91	2.3600	-78.5	0.0200	25.0	0.4	19.80
94	2.3800	-78.1	0.0200	25.0	0.4	19.72
97	2.4000	-77.8	0.0200	25.0	0.3	19.20
100	2.4200	-77.3	0.0200	25.0	0.5	19.57
104	2.4400	-76.9	0.0200	25.0	0.4	19.43
107	2.4600	-76.6	0.0200	25.0	0.3	19.71
110	2.4800	-76.2	0.0200	25.0	0.4	20.08
113	2.5000	-75.7	0.0200	25.0	0.5	20.20
116	2.5200	-75.4	0.0200	25.0	0.3	21.01
119	2.5400	-74.9	0.0200	25.0	0.5	21.38
122	2.5600	-74.5	0.0200	25.0	0.4	20.20
125	2.5800	-74.1	0.0200	25.0	0.4	19.71
128	2.6000	-73.7	0.0200	25.0	0.4	20.09
131	2.6200	-73.3	0.0200	25.0	0.4	19.14
134	2.6400	-73.0	0.0200	25.0	0.3	19.19
137	2.6600	-72.5	0.0200	25.0	0.5	20.08
140	2.6800	-72.1	0.0200	25.0	0.4	20.43
143	2.7000	-71.8	0.0200	25.0	0.3	21.29
146	2.7200	-71.3	0.0200	25.0	0.5	21.09
149	2.7400	-70.8	0.0200	25.0	0.5	20.68
152	2.7600	-70.5	0.0200	25.0	0.3	21.03
155	2.7800	-70.0	0.0200	25.0	0.5	20.80
158	2.8000	-69.7	0.0200	25.0	0.3	19.90
162	2.8200	-69.2	0.0200	25.0	0.5	20.80
165	2.8400	-68.8	0.0200	25.0	0.4	21.29
168	2.8600	-68.4	0.0200	25.0	0.4	20.71
171	2.8800	-68.0	0.0200	25.0	0.4	20.57
174	2.9000	-67.5	0.0200	25.0	0.5	19.42
177	2.9200	-67.2	0.0200	25.0	0.3	19.71
180	2.9400	-66.8	0.0200	25.0	0.4	19.73
183	2.9600	-66.4	0.0200	25.0	0.4	20.48
186	2.9800	-66.0	0.0200	25.0	0.4	21.51

Titration report

LabX

189	3.0000	-65.5	0.0200	25.0	0.5	22.87
192	3.0200	-65.1	0.0200	25.0	0.4	22.00
195	3.0400	-64.6	0.0200	25.0	0.5	20.99
198	3.0600	-64.2	0.0200	25.0	0.4	20.21
201	3.0800	-63.8	0.0200	25.0	0.4	20.73
204	3.1000	-63.5	0.0200	25.0	0.3	21.22
207	3.1200	-63.0	0.0200	25.0	0.5	22.06
210	3.1400	-62.5	0.0200	25.0	0.5	22.00
213	3.1600	-62.0	0.0200	25.0	0.5	22.28
216	3.1800	-61.7	0.0200	25.0	0.3	21.24
220	3.2000	-61.2	0.0200	25.0	0.5	20.98
223	3.2200	-60.9	0.0200	25.0	0.3	21.22
226	3.2400	-60.3	0.0200	25.0	0.6	21.70
229	3.2600	-60.0	0.0200	25.0	0.3	22.27
232	3.2800	-59.4	0.0200	25.0	0.6	22.53
235	3.3000	-59.1	0.0200	25.0	0.3	22.99
238	3.3200	-58.6	0.0200	25.0	0.5	22.72
241	3.3400	-58.1	0.0200	25.0	0.5	23.85
244	3.3600	-57.6	0.0200	25.0	0.5	23.44
247	3.3800	-57.2	0.0200	25.0	0.4	23.78
250	3.4000	-56.7	0.0200	25.0	0.5	23.29
253	3.4200	-56.2	0.0200	25.0	0.5	23.21
256	3.4400	-55.8	0.0200	25.0	0.4	23.21
259	3.4600	-55.3	0.0200	25.0	0.5	23.29
262	3.4800	-54.8	0.0200	25.0	0.5	23.20
265	3.5000	-54.4	0.0200	25.0	0.4	23.72
268	3.5200	-53.9	0.0200	25.0	0.5	24.58
271	3.5400	-53.4	0.0200	25.0	0.5	24.20
274	3.5600	-52.9	0.0200	25.0	0.5	24.49
278	3.5800	-52.4	0.0200	25.0	0.5	24.80
281	3.6000	-51.9	0.0200	25.0	0.5	24.42
284	3.6200	-51.5	0.0200	25.0	0.4	24.99
287	3.6400	-50.9	0.0200	25.0	0.6	25.80
290	3.6600	-50.4	0.0200	25.0	0.5	26.20
293	3.6800	-49.9	0.0200	25.0	0.5	26.51
296	3.7000	-49.3	0.0200	25.0	0.6	27.08
299	3.7200	-48.8	0.0200	25.0	0.5	27.22
302	3.7400	-48.3	0.0200	25.0	0.5	27.21
305	3.7600	-47.7	0.0200	25.0	0.6	27.56
308	3.7800	-47.1	0.0200	25.0	0.6	28.02
311	3.8000	-46.6	0.0200	25.0	0.5	27.93
314	3.8200	-46.1	0.0200	25.0	0.5	27.78
317	3.8400	-45.4	0.0200	25.0	0.7	28.00
320	3.8600	-44.9	0.0200	25.0	0.5	28.92
323	3.8800	-44.4	0.0200	25.0	0.5	29.52
326	3.9000	-43.7	0.0200	25.0	0.7	30.06
329	3.9200	-43.1	0.0200	25.0	0.6	30.71
332	3.9400	-42.5	0.0200	25.0	0.6	31.81
336	3.9600	-41.9	0.0200	25.0	0.6	30.99
339	3.9800	-41.2	0.0200	25.0	0.7	30.98
342	4.0000	-40.6	0.0200	25.0	0.6	31.44
345	4.0200	-40.0	0.0200	25.0	0.6	31.92

Titration report

LabX

348	4.0400	-39.4	0.0200	25.0	0.6	33.58	
351	4.0600	-38.6	0.0200	25.0	0.8	35.30	
354	4.0800	-38.0	0.0200	25.0	0.6	35.99	
357	4.1000	-37.2	0.0200	25.0	0.8	35.57	
360	4.1200	-36.4	0.0200	25.0	0.8	35.28	
363	4.1400	-35.8	0.0200	25.0	0.6	34.58	
366	4.1600	-35.2	0.0200	25.0	0.6	35.78	
369	4.1800	-34.4	0.0200	25.0	0.8	36.21	
372	4.2000	-33.6	0.0200	25.0	0.8	38.19	
375	4.2200	-32.9	0.0200	25.0	0.7	39.81	
378	4.2400	-32.0	0.0200	25.0	0.9	40.29	
381	4.2600	-31.3	0.0200	25.0	0.7	40.70	
384	4.2800	-30.4	0.0200	25.0	0.9	42.02	
387	4.3000	-29.6	0.0200	25.0	0.8	42.98	
390	4.3200	-28.7	0.0200	25.0	0.9	43.72	
394	4.3400	-27.8	0.0200	25.0	0.9	44.99	
397	4.3600	-26.9	0.0200	25.0	0.9	46.21	
400	4.3800	-26.0	0.0200	25.0	0.9	48.01	
403	4.4000	-25.0	0.0200	25.0	1.0	49.49	
406	4.4200	-24.0	0.0200	25.0	1.0	51.00	
409	4.4400	-22.9	0.0200	25.0	1.1	52.50	
412	4.4600	-21.9	0.0200	25.0	1.0	53.71	
415	4.4800	-20.8	0.0200	25.0	1.1	55.21	
418	4.5000	-19.7	0.0200	25.0	1.1	56.91	
421	4.5200	-18.5	0.0200	25.0	1.2	59.70	
424	4.5400	-17.3	0.0200	25.0	1.2	63.01	
427	4.5600	-16.0	0.0200	25.0	1.3	65.42	
430	4.5800	-14.7	0.0200	25.0	1.3	68.48	
433	4.6000	-13.2	0.0200	25.0	1.5	72.01	
436	4.6200	-11.8	0.0200	25.0	1.4	75.13	
439	4.6400	-10.3	0.0200	25.0	1.5	78.41	
442	4.6600	-8.6	0.0200	25.0	1.7	82.70	
445	4.6800	-6.9	0.0200	25.0	1.7	87.26	
448	4.7000	-5.2	0.0200	25.0	1.7	93.92	
452	4.7200	-3.2	0.0200	25.0	2.0	100.42	
455	4.7400	-1.1	0.0200	25.0	2.1	107.23	
458	4.7600	1.1	0.0200	25.0	2.2	114.96	
461	4.7800	3.6	0.0200	25.0	2.5	124.45	
464	4.8000	6.1	0.0200	25.0	2.5	132.23	
467	4.8200	8.9	0.0200	25.0	2.8	149.77	
470	4.8400	12.1	0.0200	25.0	3.2	168.97	
473	4.8600	15.8	0.0200	25.0	3.7	199.36	
476	4.8800	19.8	0.0200	25.0	4.0	218.98	
479	4.9000	25.6	0.0200	25.0	5.8	259.07	
482	4.9200	31.1	0.0200	25.0	5.5	301.38	
485	4.9400	39.1	0.0200	25.0	8.0	484.59	
488	4.9600	47.4	0.0200	25.0	8.3	767.11	
491	4.9800	64.0	0.0200	25.0	16.6	1034.43	
495	5.0000	84.0	0.0200	25.0	20.0	1193.53	
	5.007864	98.5				1198.02	EQP
502	5.0200	121.0	0.0200	25.0	37.0	1178.82	
506	5.0400	143.2	0.0200	25.0	22.2	1003.02	

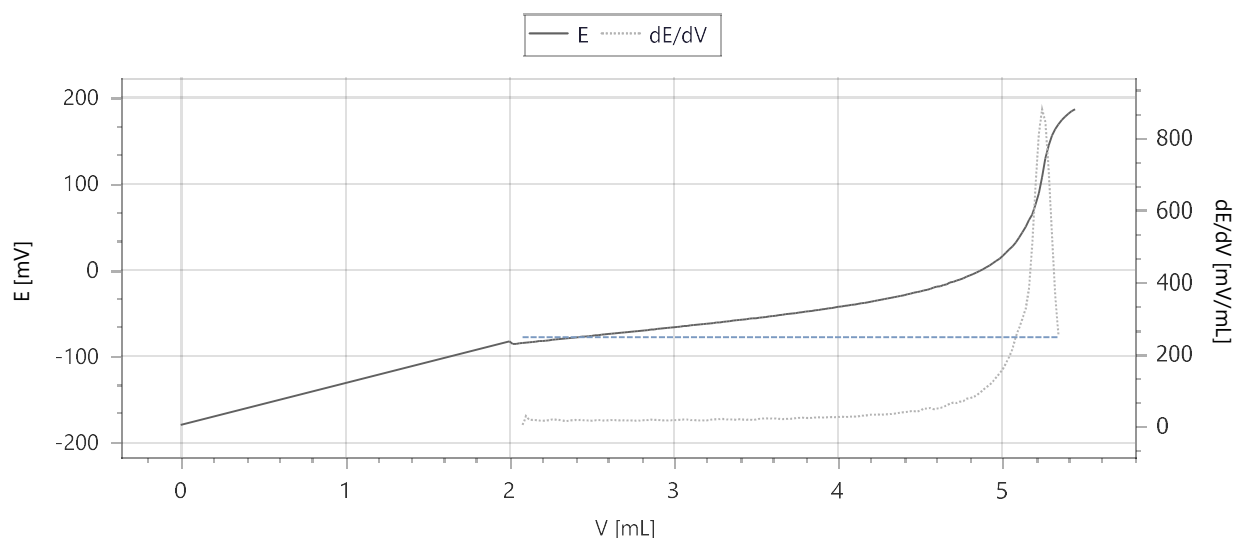
Titration report

LabX

509	5.0600	155.9	0.0200	25.0	12.7	716.06
512	5.0800	164.7	0.0200	25.0	8.8	438.12
516	5.1000	170.9	0.0200	25.0	6.2	244.26
518	5.1200	176.2	0.0200	25.0	5.3	
522	5.1400	180.4	0.0200	25.0	4.2	
525	5.1600	184.1	0.0200	25.0	3.7	
528	5.1800	187.2	0.0200	25.0	3.1	
531	5.2000	190.0	0.0200	25.0	2.8	

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-180.5		25.0			
6	2.0000	-83.6	2.0000	25.0	96.9		
40	2.0200	-86.6	0.0200	25.0	-3.0		
43	2.0400	-86.3	0.0200	25.0	0.3		
46	2.0600	-85.8	0.0200	25.0	0.5		
49	2.0800	-85.4	0.0200	25.0	0.4	6.73	
52	2.1000	-85.1	0.0200	25.0	0.3	31.52	
55	2.1200	-84.6	0.0200	25.0	0.5	21.29	
58	2.1400	-84.2	0.0200	25.0	0.4	20.75	
61	2.1600	-83.7	0.0200	25.0	0.5	20.03	
64	2.1800	-83.3	0.0200	25.0	0.4	19.19	
67	2.2000	-83.1	0.0200	25.0	0.2	18.15	
70	2.2200	-82.6	0.0200	25.0	0.5	19.28	
73	2.2400	-82.3	0.0200	25.0	0.3	19.91	
76	2.2600	-81.8	0.0200	25.0	0.5	21.88	
79	2.2800	-81.4	0.0200	25.0	0.4	21.58	
82	2.3000	-80.9	0.0200	25.0	0.5	20.78	
85	2.3200	-80.6	0.0200	25.0	0.3	19.23	
88	2.3400	-80.1	0.0200	25.0	0.5	17.61	

Titration report

LabX

92	2.3600	-79.9	0.0200	25.0	0.2	18.50
94	2.3800	-79.4	0.0200	25.0	0.5	19.75
98	2.4000	-79.2	0.0200	25.0	0.2	20.76
101	2.4200	-78.6	0.0200	25.0	0.6	20.29
104	2.4400	-78.1	0.0200	25.0	0.5	20.84
107	2.4600	-77.8	0.0200	25.0	0.3	19.39
110	2.4800	-77.5	0.0200	25.0	0.3	19.57
113	2.5000	-77.1	0.0200	25.0	0.4	18.71
116	2.5200	-76.6	0.0200	25.0	0.5	19.69
119	2.5400	-76.3	0.0200	25.0	0.3	20.30
122	2.5600	-75.8	0.0200	25.0	0.5	19.80
125	2.5800	-75.5	0.0200	25.0	0.3	18.90
128	2.6000	-75.1	0.0200	25.0	0.4	19.80
131	2.6200	-74.7	0.0200	25.0	0.4	20.01
134	2.6400	-74.3	0.0200	25.0	0.4	19.98
137	2.6600	-73.9	0.0200	25.0	0.4	19.80
140	2.6800	-73.4	0.0200	25.0	0.5	19.20
143	2.7000	-73.2	0.0200	25.0	0.2	19.81
146	2.7200	-72.7	0.0200	25.0	0.5	19.23
150	2.7400	-72.4	0.0200	25.0	0.3	19.24
152	2.7600	-71.9	0.0200	25.0	0.5	18.45
156	2.7800	-71.5	0.0200	25.0	0.4	19.26
159	2.8000	-71.3	0.0200	25.0	0.2	18.68
162	2.8200	-70.8	0.0200	25.0	0.5	20.06
165	2.8400	-70.5	0.0200	25.0	0.3	20.20
168	2.8600	-69.9	0.0200	25.0	0.6	21.58
171	2.8800	-69.6	0.0200	25.0	0.3	21.29
174	2.9000	-69.1	0.0200	25.0	0.5	20.51
177	2.9200	-68.8	0.0200	25.0	0.3	20.29
180	2.9400	-68.3	0.0200	25.0	0.5	19.42
183	2.9600	-68.0	0.0200	25.0	0.3	19.19
186	2.9800	-67.5	0.0200	25.0	0.5	19.51
189	3.0000	-67.2	0.0200	25.0	0.3	20.51
192	3.0200	-66.8	0.0200	25.0	0.4	20.42
195	3.0400	-66.4	0.0200	25.0	0.4	21.34
198	3.0600	-65.8	0.0200	25.0	0.6	21.80
201	3.0800	-65.5	0.0200	25.0	0.3	22.01
204	3.1000	-65.1	0.0200	25.0	0.4	20.51
208	3.1200	-64.6	0.0200	25.0	0.5	20.05
210	3.1400	-64.2	0.0200	25.0	0.4	19.65
214	3.1600	-63.9	0.0200	25.0	0.3	19.42
217	3.1800	-63.5	0.0200	25.0	0.4	19.56
220	3.2000	-63.0	0.0200	25.0	0.5	19.64
223	3.2200	-62.7	0.0200	25.0	0.3	21.19
226	3.2400	-62.3	0.0200	25.0	0.4	22.40
229	3.2600	-61.7	0.0200	25.0	0.6	22.48
232	3.2800	-61.4	0.0200	25.0	0.3	22.99
235	3.3000	-60.8	0.0200	25.0	0.6	23.40
238	3.3200	-60.4	0.0200	25.0	0.4	22.18
241	3.3400	-60.0	0.0200	25.0	0.4	21.73
244	3.3600	-59.5	0.0200	25.0	0.5	21.79
247	3.3800	-59.1	0.0200	25.0	0.4	21.41

Titration report

LabX

250	3.4000	-58.7	0.0200	25.0	0.4	22.60
253	3.4200	-58.2	0.0200	25.0	0.5	21.70
256	3.4400	-57.8	0.0200	25.0	0.4	21.47
259	3.4600	-57.3	0.0200	25.0	0.5	21.53
262	3.4800	-56.9	0.0200	25.0	0.4	21.12
265	3.5000	-56.6	0.0200	25.0	0.3	21.71
268	3.5200	-56.0	0.0200	25.0	0.6	23.88
272	3.5400	-55.6	0.0200	25.0	0.4	24.42
275	3.5600	-55.1	0.0200	25.0	0.5	25.07
278	3.5800	-54.5	0.0200	25.0	0.6	24.86
281	3.6000	-54.0	0.0200	25.0	0.5	24.43
284	3.6200	-53.7	0.0200	25.0	0.3	24.81
287	3.6400	-53.1	0.0200	25.0	0.6	23.49
290	3.6600	-52.6	0.0200	25.0	0.5	23.17
293	3.6800	-52.1	0.0200	25.0	0.5	23.46
296	3.7000	-51.7	0.0200	25.0	0.4	24.00
299	3.7200	-51.3	0.0200	25.0	0.4	24.22
302	3.7400	-50.7	0.0200	25.0	0.6	25.78
305	3.7600	-50.2	0.0200	25.0	0.5	26.49
308	3.7800	-49.6	0.0200	25.0	0.6	27.09
311	3.8000	-49.1	0.0200	25.0	0.5	26.50
314	3.8200	-48.6	0.0200	25.0	0.5	25.92
317	3.8400	-48.1	0.0200	25.0	0.5	26.79
320	3.8600	-47.5	0.0200	25.0	0.6	27.51
323	3.8800	-47.0	0.0200	25.0	0.5	27.99
326	3.9000	-46.4	0.0200	25.0	0.6	27.50
330	3.9200	-45.8	0.0200	25.0	0.6	27.78
333	3.9400	-45.3	0.0200	25.0	0.5	27.95
336	3.9600	-44.8	0.0200	25.0	0.5	28.78
339	3.9800	-44.2	0.0200	25.0	0.6	28.49
342	4.0000	-43.5	0.0200	25.0	0.7	28.98
345	4.0200	-43.0	0.0200	25.0	0.5	29.81
348	4.0400	-42.4	0.0200	25.0	0.6	28.94
351	4.0600	-41.9	0.0200	25.0	0.5	29.25
354	4.0800	-41.2	0.0200	25.0	0.7	30.03
357	4.1000	-40.6	0.0200	25.0	0.6	30.11
360	4.1200	-40.1	0.0200	25.0	0.5	30.50
363	4.1400	-39.3	0.0200	25.0	0.8	32.39
366	4.1600	-38.8	0.0200	25.0	0.5	32.55
369	4.1800	-38.2	0.0200	25.0	0.6	34.07
372	4.2000	-37.3	0.0200	25.0	0.9	35.44
375	4.2200	-36.6	0.0200	25.0	0.7	35.35
378	4.2400	-36.1	0.0200	25.0	0.5	35.80
381	4.2600	-35.2	0.0200	25.0	0.9	35.07
384	4.2800	-34.5	0.0200	25.0	0.7	35.90
388	4.3000	-33.9	0.0200	25.0	0.6	36.68
391	4.3200	-33.1	0.0200	25.0	0.8	37.03
394	4.3400	-32.3	0.0200	25.0	0.8	38.00
397	4.3600	-31.5	0.0200	25.0	0.8	39.37
400	4.3800	-30.9	0.0200	25.0	0.6	41.05
403	4.4000	-29.9	0.0200	25.0	1.0	42.10
406	4.4200	-29.0	0.0200	25.0	0.9	43.39

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LabX

409	4.4400	-28.3	0.0200	25.0	0.7	44.81	
412	4.4600	-27.2	0.0200	25.0	1.1	45.08	
415	4.4800	-26.4	0.0200	25.0	0.8	43.93	
418	4.5000	-25.5	0.0200	25.0	0.9	46.73	
421	4.5200	-24.5	0.0200	25.0	1.0	52.64	
424	4.5400	-23.6	0.0200	25.0	0.9	52.01	
427	4.5600	-22.5	0.0200	25.0	1.1	53.99	
430	4.5800	-21.1	0.0200	25.0	1.4	50.38	
433	4.6000	-20.0	0.0200	25.0	1.1	52.00	
436	4.6200	-19.7	0.0200	25.0	0.3	53.99	
439	4.6400	-18.1	0.0200	25.0	1.6	56.93	
442	4.6600	-17.3	0.0200	25.0	0.8	61.11	
446	4.6800	-15.3	0.0200	25.0	2.0	66.15	
449	4.7000	-14.4	0.0200	25.0	0.9	68.61	
452	4.7200	-13.0	0.0200	25.0	1.4	67.66	
455	4.7400	-11.7	0.0200	25.0	1.3	71.29	
458	4.7600	-10.2	0.0200	25.0	1.5	72.69	
461	4.7800	-8.6	0.0200	25.0	1.6	79.43	
464	4.8000	-7.0	0.0200	25.0	1.6	80.57	
467	4.8200	-5.4	0.0200	25.0	1.6	83.82	
470	4.8400	-3.7	0.0200	25.0	1.7	88.64	
473	4.8600	-1.8	0.0200	25.0	1.9	95.85	
476	4.8800	0.0	0.0200	25.0	1.8	103.33	
479	4.9000	2.3	0.0200	25.0	2.3	111.27	
482	4.9200	4.7	0.0200	25.0	2.4	117.44	
485	4.9400	7.1	0.0200	25.0	2.4	126.75	
488	4.9600	9.6	0.0200	25.0	2.5	138.48	
491	4.9800	12.5	0.0200	25.0	2.9	149.08	
494	5.0000	15.5	0.0200	25.0	3.0	160.65	
497	5.0200	19.3	0.0200	25.0	3.8	177.72	
500	5.0400	22.9	0.0200	25.0	3.6	195.45	
504	5.0600	26.7	0.0200	25.0	3.8	218.18	
507	5.0800	31.3	0.0200	25.0	4.6	252.02	
510	5.1000	37.0	0.0200	25.0	5.7	275.07	
514	5.1200	42.6	0.0200	25.0	5.6	300.02	
516	5.1400	49.6	0.0200	25.0	7.0	328.02	
520	5.1600	57.1	0.0200	25.0	7.5	383.77	
523	5.1800	63.7	0.0200	25.0	6.6	506.52	
526	5.2000	74.8	0.0200	25.0	11.1	668.96	
529	5.2200	88.5	0.0200	25.0	13.7	811.37	
533	5.2400	107.1	0.0200	25.0	18.6	881.93	
	5.242328	109.6				882.02	EQP
538	5.2600	128.5	0.0200	25.0	21.4	846.28	
541	5.2800	144.2	0.0200	25.0	15.7	711.88	
544	5.3000	155.2	0.0200	25.0	11.0	535.41	
547	5.3200	162.6	0.0200	25.0	7.4	364.80	
550	5.3400	168.4	0.0200	25.0	5.8	252.77	
553	5.3600	173.4	0.0200	25.0	5.0		
556	5.3800	177.1	0.0200	25.0	3.7		
559	5.4000	180.5	0.0200	25.0	3.4		
562	5.4200	183.4	0.0200	25.0	2.9		
565	5.4400	185.9	0.0200	25.0	2.5		

Titration report

LabX

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 INC=0.02 mL, VDISP = 2 mL	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 INC=0.02 mL, VDISP = 2 mL	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	02.03.2026 08:22:36
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	02.03.2026 08:22:36

Titration report



0.1 M TRIS - 5 mL Comment
in 60 mL H₂O

S1

String

Unused DG111
INC=0.02 mL,
VDISP = 2 mL

02.03.2026
08:22:36

Used Instruments

Name

T9_MSG_SA_002

Serial number

B551866924

Responsible User

Time

Login name

Full name

02.03.2026 08:22:36

PredefinedUser

Administrator